

Aparna Patel

Software Engineer

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Profile

High-performing **Software Engineer** with **4+ years** of experience designing scalable **microservices** for enterprise **energy domain** systems. Strong expertise in **event-driven architecture** using **RabbitMQ**, performance optimization, and **modernizing legacy Spring MVC** applications to microservices. Experienced in building **React operational dashboards** and handling production support. Hands-on with AI integration using **Spring AI** and **Azure OpenAI**, delivering intelligent incident management in production.

Technical Skills

Languages: Java, JavaScript

Backend: Spring Boot, Spring MVC, Spring Cloud (Eureka, API Gateway), REST APIs, JUnit5, Microservices, Spring AI

Messaging & Eventing: RabbitMQ, Kafka, Event-Driven Architecture

Frontend: React.js, PrimeFaces (JSF), HTML, CSS

Databases: SQL, MongoDB, VectorDB

Concepts: OOP, SOLID Principles, Multi-Threading, Data Structures & Algorithms, RAG

Tools & Platforms: Jenkins, Git, Postman, ServiceNow, Azure, AWS, Docker, JasperReports, OpenAI, Gemini

Professional Experience

CGI INDIA, Software Engineer

06/2022 – Present | Bengaluru, India

Project: Ovintiv, Canada (05/2024-Present)

- Architected **event-driven microservices** using **Spring Boot** and **Azure Service Bus**, enabling async workflows for Travel Replacement and Plan Notification modules across an oil & gas operations platform.
- Reduced incident response time by **70%** by engineering a real-time **RabbitMQ notification system** delivering ITSM alerts through a **React subscription UI**.
- Accelerated critical API response times by **64%** by implementing **parallel processing** with **CompletableFuture** and **Spring @Async**, eliminating latency bottlenecks in high-traffic endpoints.
- Designed **centralized exception handling** framework across **8+ microservices**, cutting API error inconsistencies by **40%** and significantly improving system reliability.
- Increased **unit** and **integration test** coverage across service and controller layers, reducing regression defects and improving release confidence for critical microservice workflows.
- Built a **React operations dashboard** for real-time KPI monitoring and incident analytics, eliminating manual reporting and enabling data-driven decision-making for operations teams.
- Led cross-functional root cause analysis across **10+ microservices** using log analysis, performance profiling, and collaborative debugging with AMS and onshore teams, reducing recurring production incidents by **35%**.
- Developed an **AI-powered help desk** integrating **Spring AI** with **Azure OpenAI** to automate issue triage and resolution suggestions, delivering an intelligent incident management system.

Project: GTMS (Gas Transportation Management System), Northern Ireland and ROI (06/2022-04/2024)

- Developed and enhanced **core domain modules** including Capacity, Allocation, Scheduling, and Billing using **Java** and **Spring MVC**, delivering robust business logic for a high-volume enterprise gas transportation platform.
- Refactored billing and capacity computation logic with **ExecutorService-based parallel processing**, improving performance, maintainability, and UI response time for large-scale transactional workloads.
- Optimized complex **SQL queries** for data verification and performance tuning across Billing, Allocation, and Capacity modules, reducing query execution time by up to **50%** for high-volume transactional data.
- Migrated UI from RichFaces to **PrimeFaces** (JSF) integrated with Spring MVC, reducing page load time by **53%** and improving responsiveness by **30%**.
- Developed **Jasper Reports** for Transportation, Allocation, and Billing modules, optimizing report queries and templates to reduce report generation time by **40%**.
- Supported CI/CD pipelines and application deployments using **Jenkins** in **AWS EC2** environments, monitoring build execution, managing release activities, and ensuring smooth and reliable deployments.
- Contributed to Spring Boot and Angular **modernization POC** to migrate legacy Spring MVC application toward microservices-ready architecture, **recognized in Kaizen event** for process improvement initiative.

Education

National Institute of Technology, Jalandhar, B.tech, Mechanical Engineering
CGPA- 7.41

07/2018 – 05/2022 | Jalandhar, India

Awards

- Best Collaborative Performer Award** for demonstrating strong teamwork and cross-team collaboration.
- Recognized in Kaizen event** for contributing process improvements through Spring Boot and Angular **modernization POC**.